UNIVERSITY OF CALIFORNIA, RIVERSIDE

Joint Longitudinal-Survival Analysis of Postoperative Left Ventricular Mass Index and Mortality After Aortic Valve Replacement

Ryan Kitagawa, Linlin Liu, Nancy Lopez
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1 INTRODUCTION

1.1 CLINICAL CONTEXT

Aortic valve replacement surgery is a lifesaving treatment for aortic valve disease, a heart disease that causes 28,000 deaths per year and affects roughly 2.5 percent of adults in the US (CDC, 2024). Replacement of the aortic valve is crucial for recovery from aortic valve disease, but postoperative recovery can be highly variable. A previously identified marker of cardiac recovery after surgery is the left ventricular mass index (LVMI), which measures the mass of the left ventricle on the surface of the body. Successful surgery leads to a gradual decrease in LVMI when the heart heals. However, since heart recovery can be so variable, an increase in LVMI could occur after surgery, so repeated measurements are needed after the operation to measure the LVMI of the patient over time (Lim et al., 2008; Wenn & Zeltser, 2023).

1.2 STATISTICAL PROBLEM

Postoperative patient data often contain two types of outcomes observed over time. First, LVMI is repeatedly measured at multiple follow-up visits, generating longitudinal biomarker trajectories for each patient. Second, patients are simultaneously followed for time-to-event outcomes such as death. These two processes are observed in the same individuals and may be intrinsically related. Therefore, understanding the relationship between LVMI trajectories and survival outcomes is crucial for both clinical monitoring and risk assessment.

Traditional statistical approaches typically analyze these outcomes separately. Trajectories of longitudinal biomarkers are often analyzed using linear mixed-effects models (LMMs), while survival models such as the Cox proportional hazards (PH) models are used to analyze time-to-event outcomes. However, modeling these processes separately ignores the potential dependence between them and often can lead to inefficient or biased inference.

1.3 Objectives

To address these challenges, we apply a joint longitudinal-survival model that simultaneously analyzes LVMI trajectories and time-to-event outcomes. Such framework links the longitudinal and survival processes through shared subject-specific effects, allowing the direct estimation of the association between biomarker progression and mortality risk.

The primary objective of this study is to investigate how the postoperative trajectories of LVMI after aortic valve replacement are associated with the risk of mortality. In addition, the joint model allows us to assess whether baseline patient characteristics influence LVMI progression over time and how these factors relate to survival outcomes.

2 Data Structure

2.1 Study Design

The dataset used is from an observational study of patients who underwent aortic valve replacement surgery. A total of 256 patients were followed since their surgery date to monitor their postoperative cardiac recovery and long-term outcomes, and 988 follow-up visits were recorded.

During the visits, repeated echocardiographic measurements were collected to evaluate cardiac function and structural changes in the heart. Survival information was also recorded for each patient, including the time from surgery to death or the last observed follow-up visit. As a result, the dataset contains both repeated longitudinal measurements and time-to-event outcomes observed in the same individuals, making this dataset well suited for a joint longitudinal–survival modeling framework that simultaneously analyzes biomarker trajectories and survival outcomes.

2.2 Key Variables

For the purposes of this analysis, a few variables play an important role in defining the longitudinal and survival components of the model:

- `num` – patient identifier used to link repeated measurements to individual patients,
- `log.lvmi` – natural log transformation of LVMI at each follow-up visit,
- `time` – observed time point, with surgery date as the time origin (years),
- `fuyrs` – maximum follow-up time, with surgery date as time origin (years), and
- `status` – event indicator (1 = died, 0 = lost at follow-up).

The primary longitudinal outcome considered is `log.lvmi`. Although the original LVMI variable is available, the logarithmic transformation was chosen for two reasons. First, the transformation reduces the skewness in the distribution of LVMI, helping satisfy the normality assumptions required for LMMs. Second, `log.lvmi` does not contain missing data, so we can utilize all available follow-up measurements in the analysis.

2.3 Baseline Covariates

In addition to the longitudinal and survival measurements, several baseline clinical and demographic covariates were recorded for each patient prior to surgery. They describe patient characteristics and preoperative cardiac conditions that may influence postoperative LVMI trajectories and survival outcomes.

The baseline covariates included in this analysis are:

- `sex` – patient sex (0 = Male, 1 = Female),
- `age` – age at surgery (years),
- `bsa` – preoperative body surface area (BSA),
- `lvh` – preoperative left ventricular hypertrophy (LVH),
- `prenyha` – preoperative New York Heart Association (NYHA) classification (1 = I/II, 3 = III/IV),
- `redo` – previous cardiac surgery,
- `dm` – preoperative diabetes,
- `lv` – left ventricular ejection fraction (1 = good, 2 = moderate, 3 = poor), and
- `hs` – valve type (1 = homograft, 0 = stentless porcine).

We selected these covariates based on previous studies. For example, diabetes mellitus has been shown to adversely affect left ventricular (LV) remodeling and cardiac function in patients with aortic valve disease (Lindman et al., 2011). Clinical characteristics such as age, cardiac structural measures, and body size are also known predictors of changes in left ventricular mass over time (Lee et al., 2019; Rogers et al., 2019). Because these patient characteristics can influence both the progression of LVMI and the risk of mortality, we included them as possible confounders to account for patient heterogeneity.

3 Exploratory Data Analysis

3.1 Missing Data

Due to the observational nature of the original study, we assume that the missing data in `lvmi` is missing at random (MAR). Under this assumption, the probability that an observation is missing depends on previously observed values, not on the unobserved values themselves. This assumption is commonly adopted in longitudinal analysis and allows valid inference under likelihood-based joint modeling approaches.

We evaluated the extent of missingness across the variables used in the analysis. No missing values were observed for either `log.lvmi` or the baseline covariates.

3.2 Baseline Characteristics

Next, we examined the baseline covariates between the two survival groups. Figures 3.1 and 3.2 present boxplots of continuous variables and stacked bar plots of categorical variables by survival

status, respectively.

Patients who died were generally older at the time of surgery compared to those who were lost at follow-up. The distribution of log(LVMI) also appeared slightly higher among those who died, suggesting that higher LVMI may be associated with worse outcomes.

Other baseline covariates showed moderate differences between the survival groups. Since these variables have been identified in previous clinical studies as potential predictors of LV remodeling and survival after aortic valve replacement, they were included as covariates in the models.

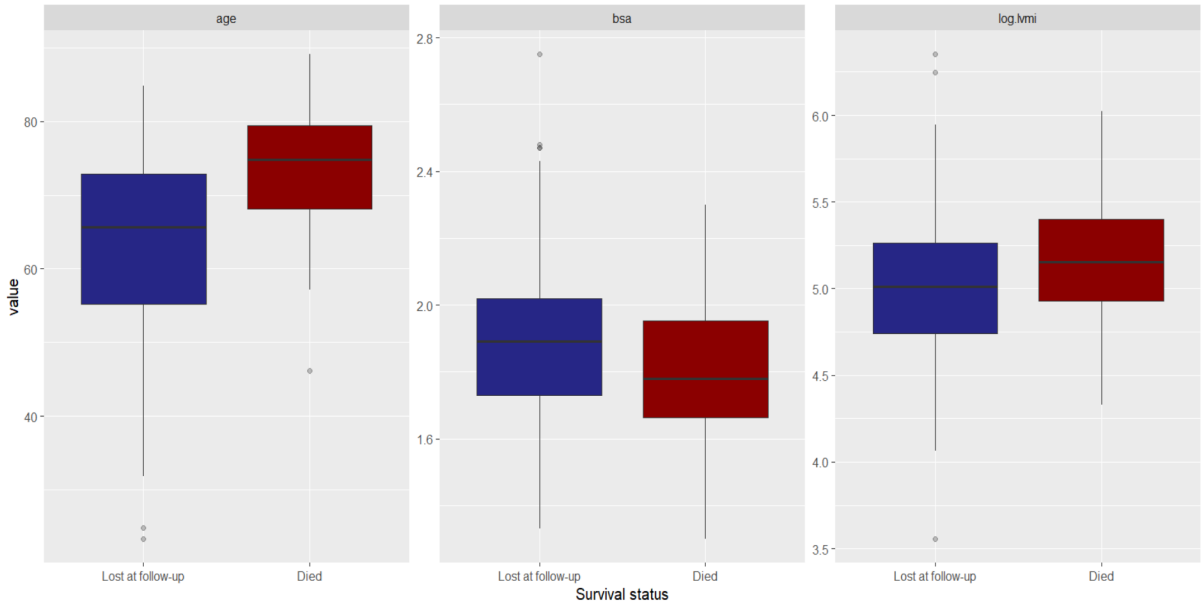

Figure 3.1: Boxplots of continuous baseline covariates by survival status.

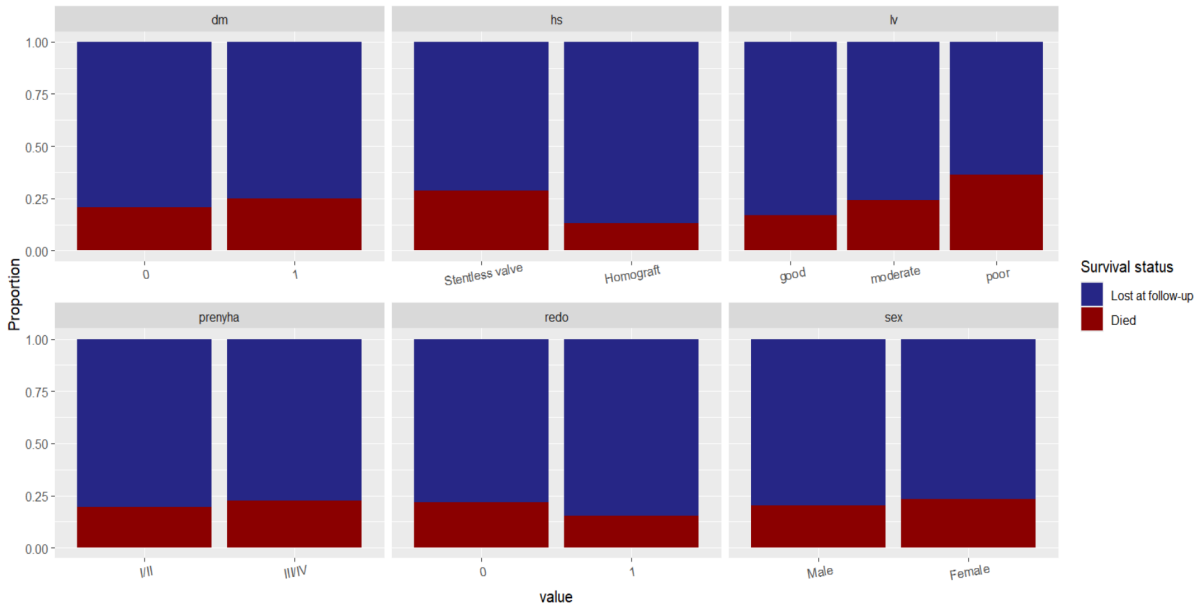

Figure 3.2: Proportions of categorical baseline covariates by survival status.

3.3 Longitudinal Trajectories of `log.lvmi`

To investigate the longitudinal change in LVMI, we plotted the individual patient trajectories of `log.lvmi` using a spaghetti plot (Figure 3.3). It reveals substantial between-patient heterogeneity, with some patients experiencing rapid decreases in LVMI after surgery and others showing more gradual changes. This variability suggests that patient-specific trajectories differ considerably and motivates the use of a LMM with random intercepts and slopes to account for subject-level heterogeneity.

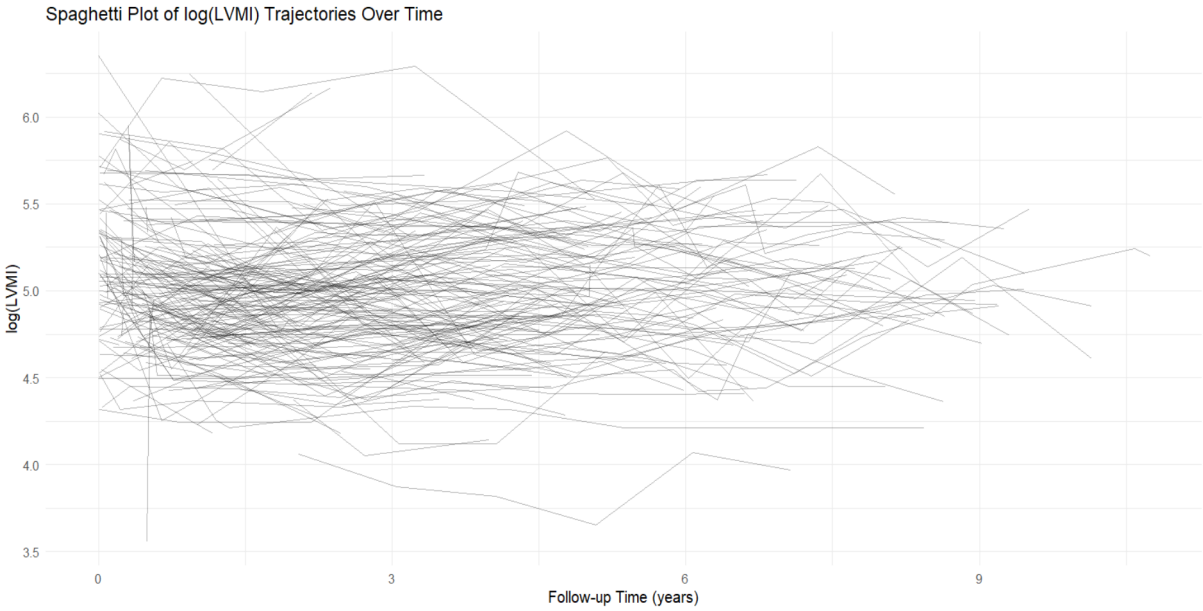

Figure 3.3: Spaghetti plot of individual patient trajectories of `log.lvmi` over time.

3.4 Survival Patterns

We further examined the overall survival outcome of the patients using a Kaplan–Meier curve in Figure 3.4. It shows a gradual decline in survival probability over the follow-up period, with most events occurring several years after surgery. Therefore, the data contain meaningful time-to-event information suitable for survival analysis.

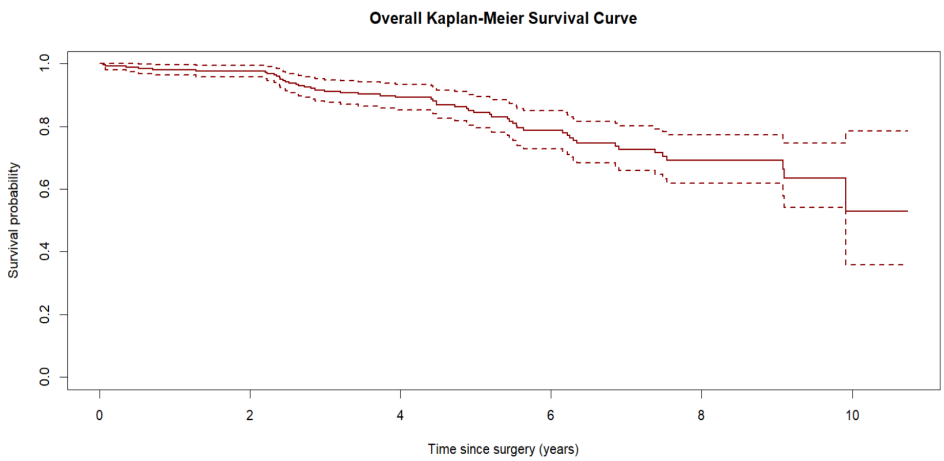

Figure 3.4: Kaplan–Meier curve of overall survival following surgery.

3.5 Average `log.lvmi` Trajectories by Survival Status

Finally, we plotted smoothed trajectories of log(LVMI by survival status (Figure 3.5) to explore the relationship between the evolution of LVMI and survival outcomes. The smoothed curves show that patients who eventually died exhibit a different trajectory pattern compared to those who were lost at follow-up. Moreover, the trajectories diverge over time. This suggests that both the level and the rate of change of LVMI may be associated with the risk of mortality. Such patterns motivate the implementation of a joint model to link the longitudinal biomarker trajectory with the survival outcome.

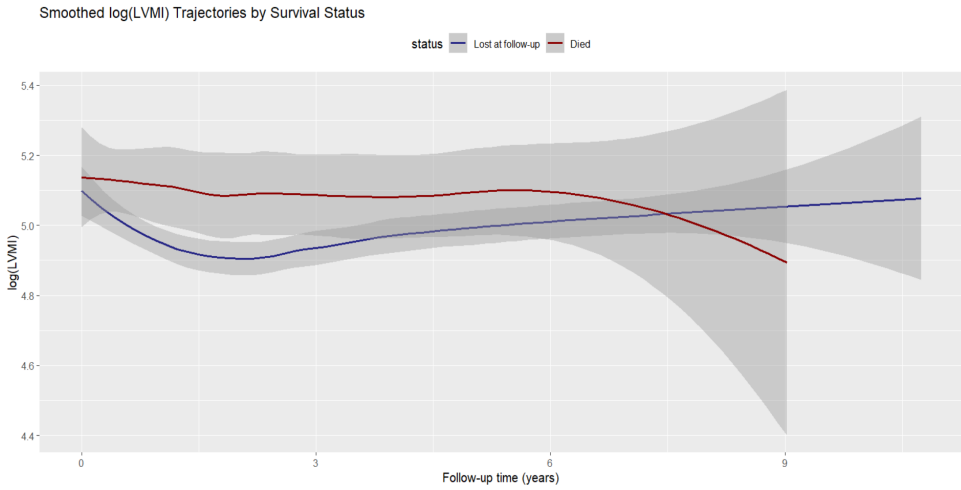

Figure 3.5: Smoothed Trajectories by Survival

Together, these EDA highlight the presence of heterogeneity in the LVMI trajectories, potential differences between survival groups, and meaningful survival information. These findings support the use of a joint modeling approach that simultaneously accounts for longitudinal biomarker evolution and time-to-event outcomes.

Our code for the EDA can be found here.

4 Methodology

4.1 Longitudinal Submodel

We modeled the longitudinal measurements of LVMI using a LMM, which allows each patient to have a unique LVMI trajectory and rate of change over time.

The model is specified as:

$$Y_{ij} = m_i(t_{ij}) + \varepsilon_{ij}, \quad \varepsilon_{ij} \overset{iid}{\sim} \mathcal{N}(0, \sigma^2),$$

where Y_{ij} represents the measured value of `log.lvmi` for subject i at time t_{ij} and is the sum of the subject-specific trajectory $m_i(t_{ij})$ and the random error ε_{ij}.

The function $m_i(t)$ represents the underlying true latent LVMI trajectory for subject i and is constructed as:

$$m_i(t) = \beta_0 + \beta_1 t + \boldsymbol{\beta}^\top \mathbf{Z}_i + b_{0i} + b_{1i} t,$$

where

- β_0 and β_1 are the intercept and slope over time for fixed effects,
- $\boldsymbol{\beta}$ represents the slope for the baseline covariates,
- $\mathbf{Z}_i$ contains the baseline covariates, and
- b_{0i} and b_{1i} represent subject-specific random intercepts and slopes, allowing subject-specific LVMI trajectories over time.

It is assumed that b_{0i} and b_{1i} follow a multivariate normal distribution:

$$\mathbf{b}_i = (b_{0i}, b_{1i})^\top \sim N(\mathbf{0}, \mathbf{D}),$$

where $\mathbf{D}$ denotes the covariance matrix of the random effects.

A LMM assumes:

- a linear relationship between the covariates and the longitudinal outcome,
- constant variance of the residual errors,
- normally distributed random effects, and
- independence between random effects and residual errors.

4.2 Survival Submodel

We fit a Cox PH survival framework to model the time-to-event outcome. This model describes how the baseline covariates affect the instantaneous risk of death at time t.

The model relies on a baseline hazard function and an exponential term involving patient covariates and is specified as:

$$h_i(t) = h_0(t) \exp(\boldsymbol{\gamma}^\top \mathbf{X}_i),$$

where

- $h_i(t)$ denotes the hazard function for patient i,
- $h_0(t)$ is the baseline hazard function, and
- $\boldsymbol{\gamma}$ represents the regression coefficients that correspond to the covariates $\mathbf{X}_{\boldsymbol{i}}$.

A Cox PH model assumes:

- constant hazard ratios for the covarites over time,
- study withdrawal is independent of the subject's prognosis, but may be influenced by the baseline covariates, and

- the functional form of the model and each covariate is correct.

4.3 Joint Model

In order to link the longitudinal outcome with the survival outcome, we combined the LMM and Cox PH survival model into a joint model. It simultaneously analyzes longitudinal measurements and time-to-event outcomes by associating the two processes through shared random effects. This framework has been widely used in biomedical studies in which repeated biomarkers are associated with clinical outcomes. By doing this, the hazard function depends on the current value of the latent LVMI trajectory.

The joint model is written as

$$h_i(t) = h_0(t) \exp\left(\boldsymbol{\gamma}^\top \mathbf{X}_i + \alpha m_i(t)\right).$$

The function $m_i(t)$ represents the latent LVMI trajectory, and α, known as the association parameter, quantifies the strength of association between the longitudinal outcome and the risk of death. This new structure links both submodels through the shared random effects in $m_i(t)$, which allows the joint model to capture subject-specific variation.

4.4 Slope Parameterization

To capture not only the level of the biomarker but also its rate of change over time, we extended the joint model by incorporating the derivative of the longitudinal trajectory into the survival model.

The new joint model becomes

$$h_i(t) = h_0(t) \exp\left(\boldsymbol{\gamma}^\top \mathbf{X}_i + \alpha_1 m_i(t) + \alpha_2 m_i'(t)\right),$$

where

- $m_i(t)$ represents the current underlying value of the LVMI trajectory for subject i,
- $m_i'(t)$ represents the instantaneous rate of change with respect to time,
- α_1 measures the association between the current level of LVMI and the hazard of death, and
- α_2 quantifies the association between the rate of change of LVMI and mortality risk.

This parameterization is motivated by both EDA and clinical evidence. Figure 3.5 shows that the trajectories of `log.lvmi` differ over time between survival groups, suggesting that not only the level of LVMI, but also the speed in which it changes over time, may be associated with mortality risk.

Previous clinical studies also suggest that the LVMI trajectories exhibit different behavior during early follow-up. In particular, Lim et al. (2008) reported that the highest rate of LVMI regression occurs within the first year after surgery, after which the rate of change slows and may even increase later in follow-up.

Therefore, including the derivative $m_i'(t)$ allows the joint model to capture potentially important dynamic features of the progression of LVMI that may provide additional prognostic information beyond the current LVMI level alone.

5 Data Analysis

5.1 Model Diagnostics

Diagnostic checks were conducted for both submodels prior to fitting the joint models. For the LMM, residual versus fitted plots were used to assess homoscedasticity. Q-Q plots of residuals and random effects were used to evaluate normality assumptions. The random intercept–slope relationship was also examined to assess the structure of the random effects. The potential nonlinearity in time was evaluated by comparing models with and without a quadratic time term. For the survival submodel, PH assumptions were assessed using Schoenfeld residual tests. Martingale residual plots were used to assess the functional form of continuous covariates. All diagnostic checks indicated that the modeling assumptions were reasonably satisfied.

5.2 Implementation

We used various packages and functions in R to implement the models previously discussed. After factorizing the categorical variables, the `lme()` function of the `nlme` package was used to model LMM. For the survival submodel, the `coxph()` function of the `survival` package was used, with `X = TRUE` to print out the design matrix from the Cox model. Finally, we utilized `JointModel()` in the `JM` package and set `method = "piecewise-PH-aGH"` to produce the joint model.

5.3 Joint Model 1: Full Model

The first joint model included all the baseline covariates in both the longitudinal and survival submodels. The shared covariates are valve type, sex, age, BSA, LVH, NYHA class, prior surgery, diabetes status, and LV ejection fraction. We included time and random intercepts and slopes for time in the LMM to account for subject-specific trajectories, and we used maximum follow-up time and clustering by patient ID in the Cox PH model.

The joint model produced a log-likelihood of -282.45, an AIC of 632.91, and a BIC of 753.44. The estimated α was about 1.25 (SE = 0.5825, $p = 0.0313$). This is initial evidence that there is a positive and statistically significant association between the underlying LVMI trajectory and the hazard of death.

5.4 Joint Model 2: Reduced Model

To obtain a simpler model, we removed the non-significant variables from the LMM. The new longitudinal model retained time, valve type, sex, age, BSA, LVH, diabetes status, and LV ejection fraction. We also performed a stepwise backward selection based on AIC on the survival submodel AIC to identify relevant predictors. This procedure retained age and LV ejection fraction as the primary predictors of survival.

The reduced joint model yielded a log-likelihood of -285.52, an AIC of 621.05, and a BIC of 709.68. The lower AIC and BIC values compared to Joint Model 1 suggest an improved model fit with the reduced specification. The estimated association parameter was $\alpha \approx 0.99$ (SE = 0.5458, $p = 0.0696$), indicating a weaker association between LVMI and survival relative to the full model.

5.5 Joint Model 3: Final Model

For the final model, we kept the same covariates for the LMM, but age and only the indicator for severe dysfunction of LV instead both levels for the Cox PH model. This joint model produced a log likelihood of -286.29, an AIC of 620.58, and a BIC of 705.66. The value of α was ≈ 1.06 with a standard error of 0.5425 and a p-value of 0.0507. This model shows a slightly stronger and marginally significant positive association than the second model and has the best fit of all three models.

The estimated association parameter indicates that higher underlying values of LVMI are associated with an increased hazard of death. Specifically, increases in the latent LVMI trajectory correspond to a higher mortality risk, so LVMI progression may serve as an important prognostic marker for patient outcomes after valve replacement surgery.

Model	Log Likelihood	AIC	BIC
Full Model	-282.45	632.91	753.44
Reduced Model	-285.52	621.05	709.68
Final Model	-286.29	620.58	705.66
Slope Model	-287.87	621.74	703.28

Table 5.1: Comparison of joint model specifications.

Variable	Estimate	SE	p-value
`time`	-0.008	0.005	0.130
`hsHomograft`	-0.071	0.045	0.120
`sexFemale`	**-0.214**	**0.049**	**<0.0001**
`age`	-0.001	0.002	0.563
`bsa`	**-0.270**	**0.097**	**0.005**
`lvh1`	-0.123	0.068	0.071
`dm1`	**-0.191**	**0.094**	**0.042**
`lv2`	-0.007	0.043	0.880
`lv3`	0.138	0.073	0.058

Table 5.2: Fixed effects estimates from the longitudinal mixed-effects model for `log.lvmi`.

Variable	Coefficient	Hazard Ratio (HR)	SE	p-value
`age`	**0.118**	**1.125**	**0.019**	**<0.0001**
`lv3`	**1.197**	**3.309**	**0.410**	**0.004**

Table 5.3: Results from the survival submodel. Hazard ratios represent multiplicative effects on the hazard of death.

Random Effect	Standard Deviation	Correlation
Intercept	0.307	
Slope (`time`)	0.042	-0.356
Residual	0.193	

Table 5.4: Random effects variance components for the longitudinal model.

Our EDA suggested differences in the LVMI trajectories between the two survival groups, particularly in early follow-up. The results of our final joint model confirm this relationship, showing a positive association between the latent trajectory of LVMI and the risk of mortality. This indicates that higher LVMI values over time are associated with an increased risk of death after surgery.

5.6 Slope Parameterization

Before refitting our final joint model to incorporate the slope parameterization, we first cleaned our dataset by removing duplicate time points within subjects to exclude measurements occurring after the event or censoring time and restricting the analysis to individuals with complete covariate information.

We then implemented the `JM()` function in the `JM` package. Instead of using the default parameterization method `value`, we set it to `slope`. The derivative of the longitudinal trajectory with respect to time was specified using the `derivForm()` argument, which identifies the fixed and random effects associated with the time variable.

This model produced an α of approximately -6.15 (SE = 6.2488, $p = 0.3247$). Thus, the LVMI change was not significantly associated with survival in this dataset. In addition, the model fit statistics were similar to those of the third joint model. These results suggest that the current value of LVMI provides a stronger association with mortality risk than the instantaneous rate of change. Therefore, Joint Model 3 with parameterization of `value` was chosen as our final model.

5.7 Sensitivity Analysis

After retaining our final joint model, we performed sensitivity analyses to ensure that our findings are robust.

The first sensitivity analysis was on the choice of the Cox PH model. To test whether our covariate selection was correct, we extended the survival submodel by adding sex and valve types as covariates. Using this updated survival submodel and the longitudinal submodel of the final model, we refit the joint model, which had an AIC of 624.33. Seeing that the extra covariates did not improve the model fit, we can conclude that our final model is robust for the survival submodel.

Furthermore, we used both the value and slope parameterization on the joint model by setting `parameterization = "both"` to test if this improves the fit. This model resulted in an AIC value of 621.74, which is about the same as the final model. Therefore, adding a slope and value does not improve the model fit and confirms that the LVMI trajectories over time are not associated with the rate of change. With this in mind, we conclude that our final model is robust and sufficient.

Our code for the data analysis can be found here.

6 Conclusion

In this study, we used a joint longitudinal-survival model to examine how the relationship between postoperative LVMI trajectories and mortality after aortic valve replacement surgery. This approach allowed us to analyze repeated LVMI measurements together with time-to-event outcomes while accounting for the association between the two processes.

The results show that LVMI trajectories are associated with the risk of mortality. Higher LVMI levels over time were linked to an increased hazard of death. Several baseline covariates were found to be significantly related to mortality, including age and LV function. On the other hands, covariates like sex, BSA, and diabetes status were associated with differences in LVMI trajectories over time.

We also explored a joint model where the hazard depended on both the current LVMI level and its rate of change. However, the slope term did not improve the model fit, suggesting that the current LVMI level captures most of the prognostic information in the dataset we used.

Sensitivity analyses were conducted to assess the robustness of our final model. Both alternative survival model specifications and joint model parameterization specifications produced similar results, supporting the overall association between LVMI trajectories and mortality risk.

Several limitations are present in this analysis. First, the data are observational, so causal conclusions cannot be made. Additionally, The results depend on the assumed model structure and the available covariates, so other clinical and physical factors not included may also influence outcomes.

Future work could extend this analysis by incorporating additional cardiac biomarkers or exploring more flexible longitudinal models. Dynamic prediction tools based on updated LVMI measurements may also provide more insight for risk assessment and patient monitoring.

7 Contribution

All members contributed equally to the timely completion of this project. We worked together, regularly reviewing each other's work and discussing and exchanging ideas. Our final project reflects the combined and collaborative effort of all team members.

Name	EDA	Primary Model	Extension Model and Sensitivity Analysis	Presentation	Report	GitHub Repository
R. K.	✓	✓		✓	✓	
L. L.			✓	✓	✓	✓
N. L.	✓	✓		✓	✓	

Table 7.1: Breakdown of member contribution for the final project.

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